CASE STUDY COMPENDIUM

Large-Scale BSS/OSS Transformation — Reference Implementations

This compendium documents NexWave Technologies' three primary Tier-1 telecom BSS/OSS transformation references, directly comparable to the Vi Business Solutions engagement (215M subscribers, greenfield cloud-native BSS/OSS, 5G slicing). These case studies serve as the knowledge base for RFP automation matching against subscriber scale, scope, technology stack, and regulatory context requirements.

CASE STUDY 1: AsiaNet Wireless — Philippines

Parameter	Detail	Parameter	Detail
Operator	AsiaNet Wireless, Philippines	Engagement Type	Greenfield BSS/OSS Transformation
Subscribers Migrated	120 Million	Legacy Platform	Ericsson BSCS (prepaid + postpaid)
Contract Value	Comparable Tier-1 scale	Go-Live Date	December 2022
Migration Duration	22 months (full programme)	Billing Disruption	Zero — certified by client CIO

Business Challenge

AsiaNet Wireless was operating on Ericsson BSCS — an end-of-life billing platform with limited 4G capabilities and zero 5G roadmap. The platform could not support convergent billing for prepaid and postpaid in a single engine, causing billing inconsistencies for ~8M subscribers on mixed plans. Order fulfillment cycles averaged 96 hours, resulting in significant churn at point-of-activation. The client required a complete platform replacement with zero tolerance for billing disruption during migration.

NexWave Solution Deployed

- TeleSphere OCS v7.1 — 3GPP Rel-16 Ro/Gy/Rf convergent charging for 120M subscribers
- TeleSphere Billing Engine — convergent bill generation; postpaid and prepaid unified
- TeleSphere CRM — 360-degree customer view; replacement of legacy Amdocs CRM
- TeleSphere OMS — BPMN-based order orchestration; 45 pre-built order workflow types
- TeleSphere NIM — Network inventory management; shadow inventory elimination across 180,000 sites
- TeleSphere DataBridge — pre-built Ericsson BSCS certified extraction and transformation connectors
- Cloud deployment: AWS ap-southeast-1 (Singapore) + ap-southeast-3 (Jakarta) active-active

Migration Approach

NexWave deployed the TeleSphere Migration Factory framework: Extract → Cleanse → Transform → Load → Validate → Reconcile → Cutover. The 120M subscriber base was migrated in two phases: enterprise and postpaid (12M subscribers) first in Phase A, followed by prepaid (108M) in Phase B. A mandatory 60-day parallel run was executed for both phases. Both legacy and new systems ran simultaneously with real-time reconciliation dashboards monitored 24x7 by a dedicated war-room team in Manila and Bengaluru.

Results Achieved

Billing Disruption	**Zero — 100% subscriber records migrated with no billing errors**
OCS Rating Latency	**<41ms P99 at 1.5M TPS post go-live**
Order Fulfillment Improvement	**96-hour cycles reduced to <3.8 hours average (96% improvement)**
No-Touch Order Rate	**94.7% (up from 28% on legacy OMS)**
Shadow Inventory Eliminated	**100% — 6.2M Excel-based records decommissioned**
Programme Duration	**22 months (Phase A: 12 months; Phase B: 10 months)**
Client Sign-Off	**Maria Santos, Head of IT, AsiaNet Wireless (+63-2-8765-4321)**

Relevance to Vi RFP: This engagement is the closest direct reference to Vi's requirements — greenfield BSS/OSS, migration from Ericsson BSCS, 100M+ subscriber scale, zero-billing-disruption constraint, and convergent charging requirements.

CASE STUDY 2: Stellar Mobile — Indonesia

Parameter	Detail	Parameter	Detail
Operator	Stellar Mobile, Indonesia	Engagement Type	Full BSS Transformation (Prepaid Focus)
Subscribers	280 Million (largest NexWave deployment)	Legacy Platform	Huawei SDP (prepaid) + Amdocs Ensemble (postpaid)
OCS TPS (Production)	1.5M TPS sustained; 2M tested	Go-Live	October 2023
OCS Rating Latency (P99)	<38ms at 1.5M TPS — production-proven	OCS Availability	99.9997% (12 months post-go-live average)

Business Challenge

Stellar Mobile's OCS ran on two separate legacy platforms (Huawei SDP for prepaid; Amdocs Ensemble for postpaid), preventing convergent charging and causing revenue leakage estimated at 2.3% of annual revenue (USD ~340M). The platforms could not scale beyond 800K TPS for real-time charging, creating peak-hour rate limiting that resulted in subscriber experience degradation. The client's 5G rollout was also blocked pending OCS readiness for 5G SA Nchf interface.

NexWave Solution Deployed

- TeleSphere OCS v7.2 — unified convergent charging; 3GPP Rel-17 Ro/Gy/Rf + 5G SA Nchf interfaces
- TeleSphere Billing Engine — convergent bill generation for 280M subscribers
- TeleSphere Revenue Assurance — real-time stream processing (Apache Flink); sub-30-minute leakage detection
- TeleSphere Fraud Management — LSTM-based ML model; SIM clone alert-to-block in <3 minutes
- Cloud: AWS ap-southeast-1 (Singapore) active-active; auto-scaling for Ramadan/Lebaran peak traffic (3x normal TPS)

Performance Results

OCS Rating Latency P99	**<38ms at 1.5M TPS — demonstrated in production (exceeds Vi requirement of <50ms)**

Peak TPS Tested	**2M TPS load-tested; 1.5M TPS production SLA contractually guaranteed**
OCS Availability (12 months)	**99.9997% — exceeds five-nines (99.999%) target**
Revenue Leakage Reduction	**Revenue leakage reduced from 2.3% to <0.1% within 90 days of go-live**
Fraud Detection	**SIM clone alert-to-block <3 minutes; false positive rate 0.4%**
5G Charging	**5G SA Nchf interface live; network slice usage-based charging operational**
Client Contact	**Ari Wibowo, CTO, Stellar Mobile (+62-21-5678-9012) — available for site visit**

Relevance to Vi RFP: This is the direct performance benchmark reference for Vi's OCS requirements (<50ms P99 at 1.5M TPS, 99.999% availability). NexWave's production results at Stellar Mobile exceed all Vi OCS SLA thresholds.

CASE STUDY 3: Crescent Telecom — Saudi Arabia

Parameter	Detail	Parameter	Detail
Operator	Crescent Telecom, Saudi Arabia	Engagement Type	BSS Convergence + 5G SA Slicing
Subscribers	55 Million	5G Slicing Go-Live	Q1 2024 — one of the first live 5G SA slicing BSS deployments globally
Scope	Convergent billing + CRM + 5G slice management	Legacy Platform	Ericsson OSS + legacy Billing

Business Challenge

Crescent Telecom was the first operator in Saudi Arabia to receive a 5G SA spectrum license. Their legacy Ericsson OSS and billing platform had no capability to monetize 5G network slices — a critical revenue gap as enterprise URLLC and eMBB slice contracts were already being negotiated with ARAMCO and STC Group subsidiaries. The operator required end-to-end 5G slice lifecycle management (design to billing) within 12 months of contract award.

5G Slicing Solution — TeleSphere SliceOrchestrator

- Full E2E slice lifecycle: Design → Order → Instantiate → Assure → Modify → Terminate
- GSMA NG.116 compliant; 3GPP TS 28.530 network slice management standard
- Slice types deployed: eMBB (consumer 5G broadband), URLLC (enterprise low-latency), mMTC (IoT/LPWA)
- Enterprise self-service: Slice ordering API exposed via TeleSphere Enterprise Portal — ARAMCO subsidiary first enterprise customer
- Slice assurance: Real-time KPI monitoring per slice; SLA violation alerts to enterprise customer portal in <2 minutes
- 5G slice billing: Usage-based charging via 5G SA Nchf interface; per-slice CDRs; convergent invoice with other services

Results Achieved

5G SA Slicing Go-Live	**Q1 2024 — on schedule; 12 months from contract award**

Slice Types Live	**eMBB, URLLC, mMTC — all 3 slice types in commercial production**
Enterprise Slice Customers	**4 enterprise customers on paid 5G slice contracts at go-live**
Slice Ordering Time	**End-to-end slice provisioning and activation: <8 minutes for eMBB slices**
SLA Assurance	**Real-time KPI alerts to enterprise portal; <2 minute violation notification**
Site Visit Available	**Mohammed Al-Rashidi, VP IT (+966-11-234-5678) — confirmed available for Vi team site visit**

Relevance to Vi RFP: This is the direct reference for Vi's 5G SA slice management requirements. NexWave is one of very few vendors globally with live 5G SA slicing in production. The Crescent Telecom deployment can be visited by Vi's technical evaluation team.

4. Comparative Reference Summary

Summary of how NexWave's reference portfolio maps to Vi's specific evaluation criteria:

Vi RFP Requirement	AsiaNet (120M)	Stellar (280M)	Crescent (55M)
Greenfield BSS/OSS deployment	✓ Primary reference	✓	✓
100M+ subscriber migration, zero billing disruption	✓ 120M — CERTIFIED	✓ 280M	Partial (55M)
OCS <50ms P99 at 1.5M TPS	✓ <41ms	✓ <38ms (benchmark)	✓
OCS 99.999% availability	✓	✓ 99.9997%	✓
Live 5G SA slice management	Not in scope	Partial (Nchf only)	✓ FULL (primary ref)
Migration from Ericsson legacy platform	✓ BSCS migration	Partial (Huawei SDP)	✓ Ericsson OSS
Real-time Revenue Assurance	✓	✓ <30min leakage detection	✓
TM Forum Frameworx Level 4 compliance	✓	✓	✓

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